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BASIC NETWORKING COMMANDS IN LINUX AND WINDOW OS.

Aim:

To study and execute basic networking commands in windows and linux OS for viewing network config, testing connectivity checking route information, monitoring network status and troubleshoot network related issues.

Commands:

1. IP CONFIG:

Displays network configuration

- Windows command - ipconfig
- LINUX command - ip addr (or) if config

2. PING:

Test connectivity to a remote host

- Windows command - ping google.com
- Linux command - ping google.com

3. Trace ROUTE:

Display the path packets take to reach a destination.

- Windows command - tracert google.com
- Linux command - traceroute google.com

4. NS LOOKUP:

Resolves domain name to IP address

- Windows command - nslookup google.com
- Linux command - nslookup google.com

```
[sanjay@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fef2:96fd  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:f2:96:fd  txqueuelen 1000  (Ethernet)
        RX packets 26  bytes 3188 (3.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 69  bytes 7363 (7.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 18  bytes 2112 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 18  bytes 2112 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[sanjay@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[sanjay@localhost ~]$ hostname
localhost.localdomain
[sanjay@localhost ~]$ nslookup google.com
Server:         10.252.249.176
Address:        10.252.249.176#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.205.14
Name:   google.com
Address: 2404:6800:4007:801::200e
[sanjay@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
MTU    OMTU Security
Proto  Destination        Source              State      Channel
[sanjay@localhost ~]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[sanjay@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
   group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
mode DEFAULT qlen 1000
    link/ether 08:00:27:f2:96:fd brd ff:ff:ff:ff:ff:ff
[sanjay@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 10.252.249.176
[sanjay@localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
   group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
mode DEFAULT qlen 1000
    link/ether 08:00:27:f2:96:fd brd ff:ff:ff:ff:ff:ff
[sanjay@localhost ~]$ ss -tuln
Netid   State    Recv-Q   Send-Q      Local Address:Port      Peer Address:Port
udp     UNCONN   0        0             0.0.0.0:5353           0.0.0.0:*
udp     UNCONN   0        0           127.0.0.1:323            0.0.0.0:*
udp     UNCONN   0        0             [::]:5353              [::]:*
udp     UNCONN   0        0             [::1]:323              [::]:*
tcp     LISTEN   0        4096        127.0.0.1:631            0.0.0.0:*
tcp     LISTEN   0        128           0.0.0.0:22             0.0.0.0:*
tcp     LISTEN   0        4096            [::1]:631              [::]:*
tcp     LISTEN   0        128             [::]:22                [::]:*
```

```
Address: 2404:6800:4007:801::200e

[sanjay@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp        0      0 localhost.localdo:57058 mirror.twds.com.t:https TIME_WAIT
tcp        0      0 localhost.localdo:52042 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:57086 mirror.twds.com.t:https TIME_WAIT
tcp        0      0 localhost.localdo:57074 mirror.twds.com.t:https TIME_WAIT
tcp        0      0 localhost.localdo:50946 250621808.mrs.cdn:https ESTABLISHED
tcp        0      0 localhost.localdo:52052 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:54010 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:52066 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:53984 ftp.yz.yamagata-u.:http TIME_WAIT
tcp        0      0 localhost.localdo:37922 mirror.twds.com.t:https TIME_WAIT
tcp        0      0 localhost.localdo:54000 ftp.yz.yamagata-u.:http TIME_WAIT
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 [::]:mdns               [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ACC ]     STREAM     LISTENING     29122    /tmp/.ICE-unix/2498
unix  2      [ ACC ]     STREAM     LISTENING     20872    /run/systemd/private
unix  2      [ ACC ]     STREAM     LISTENING     20887    /run/lvm/lvmpolld.so
cket
unix  2      [ ACC ]     SEQPACKET  LISTENING     20889    /run/systemd/coredump
p
unix  2      [ ACC ]     SEQPACKET  LISTENING     20891    /run/udev/control
unix  2      [ ACC ]     STREAM     LISTENING     29379    /tmp/.X11-unix/X0
unix  2      [ ACC ]     STREAM     LISTENING     29381    /tmp/.X11-unix/X1
```

```
    link/ether 08:00:27:f2:96:fd brd ff:ff:ff:ff:ff:ff
[sanjay@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 10.252.249.176
[sanjay@localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
   group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
gode default qlen 1000
    link/ether 08:00:27:f2:96:fd brd ff:ff:ff:ff:ff:ff
[sanjay@localhost ~]$ ss -tuln
Netid   State    Recv-Q  Send-Q          Local Address:Port       Peer Address:Port
udp     UNCONN   0       0                 0.0.0.0:5353            0.0.0.0:*
udp     UNCONN   0       0               127.0.0.1:323             0.0.0.0:*
udp     UNCONN   0       0                 [::]:5353               [::]:*
udp     UNCONN   0       0                 [::1]:323               [::]:*
tcp     LISTEN   0       4096            127.0.0.1:631             0.0.0.0:*
tcp     LISTEN   0       128               0.0.0.0:22              0.0.0.0:*
tcp     LISTEN   0       4096              [::1]:631               [::]:*
tcp     LISTEN   0       128               [::]:22                 [::]:*

[sanjay@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
[sanjay@localhost ~]$
```

5. NETSTAT:

Displays active network connections and listening ports.

Windows command - netstat / netstat -a

Linux command - netstat -a / ss -a

6. ARP:

Displays the ARP cache

Windows command - arp -a

Linux command - arp -a / ip neigh

7. HOSTNAME:

Displays system hostname

Windows command - hostname

Linux command - hostname

8. ROUTE:

Displays or modifies the routing table.

Windows command - route print

Linux command - ip route

9. MAC ADDRESS:

Display MAC address of computer

Windows command - getmac

Linux command - ip link

Result:

Thus, basic networking commands are executed in both windows CMD and linux terminal successfully.